Epistemic Determinism in Large Language Models

Epistemic Determinism and the Architecture of Causal-Topological Orchestrated Inference in Stochastic Large Language Models: Theoretical Framework, Mathematical Model, and Empirical Validation

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Abstract

Autoregressive Large Language Models (LLMs) operate as stochastic token probability generators whose unconstrained inference paths are fundamentally susceptible to hallucinations, semantic drift, error compounding, and epistemic instability. This paper presents the **Quorum** architecture: a formal neurosymbolic computational framework that transforms unconstrained stochastic generative processes into a deterministically supervised, epistemically anchored state machine. The theoretical foundation comprises:

1. Propositional epistemic atomization via the five-tuple Linked Atom Graph (LAG),
2. Wave-based topological execution over a Cognitive Directed Acyclic Graph (Cognitive DAG),
3. Topological Directed Atom (TDA) poset wave scheduling governed by an absolute lexical null hypothesis,
4. Autonomous structural argument evaluation via discrete graph surgery, Assumption-Based Truth Maintenance Systems (ATMS), and Toulmin's argumentation structure, and
5. An orthogonal two-factor BARS evaluation matrix combining asymmetric Popperian falsification with continuous piecewise power-curve damping.

We formalize this framework through two core architectural principles (confluent wave scheduling under Kahn partial ordering and epistemic warrant reachability under discrete ATMS graph surgery) and two mathematically verified system properties (deterministic lexical error suppression and monotonic Lipschitz score stability). The framework is empirically validated through a strategic foresight case study (Sitra Megatrend Audit, $N = 255$ causal atoms), demonstrating how the system penalizes passive conversational delegation (HACM: 11.7%, Causal Score: 7.2%) and detects pseudo-competence despite clean linguistic presentation (jargon load $0.00/2.0$ vs. cognitive depth $1.47/3.0$). Finally, we synthesize Quorum's positioning across the landscape of reasoning architectures, retrieval frameworks, and cognitive governance paradigms.

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1. Introduction and Problem Formulation

An autoregressive language model is formally defined as a conditional probability distribution over a discrete vocabulary $\mathcal{V}$:

$$P(Y \mid X; \theta) = \prod_{t=1}^{T} P(y_t \mid y_{<t}, X; \theta), \quad y_t \in \mathcal{V} \tag{1}$$

where $X = (x_1, \ldots, x_N)$ is an input prompt sequence and $\theta \in \mathbb{R}^D$ represents the static model weights. Although modern frontier models produce syntactically fluent discourse, their epistemic utility in high-stakes reasoning is bounded by three structural failures:

1. **Aleatoric Uncertainty and Calibration Divergence:** The distribution $P(Y \mid X; \theta)$ models statistical token co-occurrence frequencies across training corpora rather than objective truth values ($\mathcal{T} \neq \mathcal{P}$). The model generates statistically plausible assertions that lack empirical backing.
2. **Conflation of Association with Causation:** Autoregressive attention captures symmetric correlation $P(A \mid B)$ but cannot distinguish a genuine causal mechanism $B \to A$ from confounding background factors ($C \to A \land C \to B$) or reverse causation ($A \to B$).
3. **Superlinear Error Cascade:** In unconstrained multi-step inference, a minute semantic drift at step t conditions the distribution at step $t+1$, causing errors to compound exponentially across generative trajectories:

$$\mathbb{E}\left[\|\hat{y}_{t+k} - y^*_{t+k}\|\right] \sim \mathcal{O}\left(e^{\lambda k}\right), \quad \lambda > 0 \tag{2}$$

The Quorum architecture addresses these vulnerabilities by stripping the LLM of autonomous state transitions. The generative model is restricted to a stateless, constrained propositional sensor, while the control logic, state progression, and evidentiary evaluation are enforced by an external, mathematically validated neurosymbolic engine.

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2. Epistemic Atomization and the Linked Atom Graph (LAG)

2.1. Atomization Morphism

Let an unstructured natural language input or discourse history be denoted as $\mathcal{D} \in \Sigma^*$. We define the atomization morphism $\alpha : \Sigma^* \to \mathcal{A}$ that maps the raw discourse into a finite set of minimal, mutually independent propositional assertions:

$$\mathcal{A} = \{a_1, a_2, \dots, a_n\} \tag{3}$$

Every epistemic atom $a_i \in \mathcal{A}$ is formally defined as an immutable five-tuple:

$$a_i = \langle \tau_i, \mathcal{C}_i, \sigma_i, \pi_i, \kappa_i \rangle \tag{4}$$

- $\tau_i \in \mathcal{L}_{\text{prop}}$ is an atomic proposition formulated in formal logic with unambiguous truth-evaluable acceptance criteria.
- $\mathcal{C}_i \subset \Sigma^*$ is the spatial and contextual bounding box isolating the exact text span in $\mathcal{D}$ from which τ_i was derived.
- $\sigma_i \in \{\text{USER}, \text{ASSISTANT}\}$ is the speaker attribution index, separating human supervisory intent from model-generated synthetic drafts.
- $\pi_i \in \{+1, -1\}$ is the evidentiary polarity index: $\pi_i = +1$ denotes a positive capability claim (demonstrating analytical competence), whereas $\pi_i = -1$ designates an asymmetric anti-pattern detector (identifying cognitive failure modes, fallacies, or uncritical adoption).
- $\kappa_i \in \{0, 1\}$ is the lexical provenance anchoring index, governed by an uncompromising null hypothesis:

$$\kappa_i = \begin{cases} 1, & \text{find}(\text{Norm}(\mathcal{C}_i), \text{Norm}(\mathcal{D})) \neq -1 \ \wedge \ \text{Prov}(\mathcal{C}_i, \sigma_i) \ \wedge \ \text{Triangulate}(\mathcal{T}_i, \mathcal{C}_i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\text{Norm}(\cdot)$ denotes Unicode NFKC diacritic decomposition and formatting strip, $\text{Prov}(\cdot)$ enforces structured XML speaker isolation, and $\text{Triangulate}(\cdot)$ validates internal CoT consistency against syntactic anchors.

The Syntactic Provenance vs. Semantic Entailment Boundary: An exact substring match $\text{find}(\mathcal{C}_i, \mathcal{D}) \neq -1$ provides *physical provenance*—irrefutable proof that citation $\mathcal{C}_i$ was not fabricated or spliced across disparate paragraphs. However, deterministic string matching is semantically blind: it proves the physical existence of characters, not that proposition τ_i logically follows from $\mathcal{C}_i$. In Quorum, this semantic gap is bridged without falling into second-order LLM stochasticity because **propositions τ_i are NOT freely invented by the model**. Instead, propositions are **pre-compiled, strictly typed evaluation criteria** defined in Single Source of Truth (SSOT) Behaviorally Anchored Rating Scale (BARS) rubrics (`seed_data.json`). The generative model is constrained to identifying the exact verbatim clause $\mathcal{C}_i$ satisfying the pre-defined target invariant τ_i, backed by reasoning trace syntactic anchor triangulation and asymmetric Popperian falsification ($\pi_i = -1$). If physical containment or provenance fails in deterministic Python code, $\kappa_i = 0$ and the proposition fails closed unconditionally.

2.2. Linked Atom Graph Structure and Root Node Typology

The extracted atoms are structured into a directed, attributed multi-graph $\mathcal{G}_{\text{LAG}} = (\mathcal{A}, \mathcal{E}_{\text{dep}}, \mathcal{E}_{\text{temp}}, \mathcal{E}_{\text{causal}})$, where edge sets define logical dependency, temporal order, and inferred causal transmission.

To avoid misclassifying legitimate foundational axioms, $\mathcal{G}_{\text{LAG}}$ partitions root nodes ($\text{InDegree}(a_i) = 0$) into two distinct epistemic classes:

1. **Exogenous Boundary Axioms ($\mathcal{A}_{\text{axiom}}$):** Primitive operational constraints or historical baseline premises (U_i) that require no internal textual ancestors (e.g., establishing the scope of an external institutional report).
2. **Unjustified Orphan Claims ($\mathcal{A}_{\text{orphan}}$):** Dependent inferential conclusions asserted without supporting evidence, warrants, or mediating causal links.

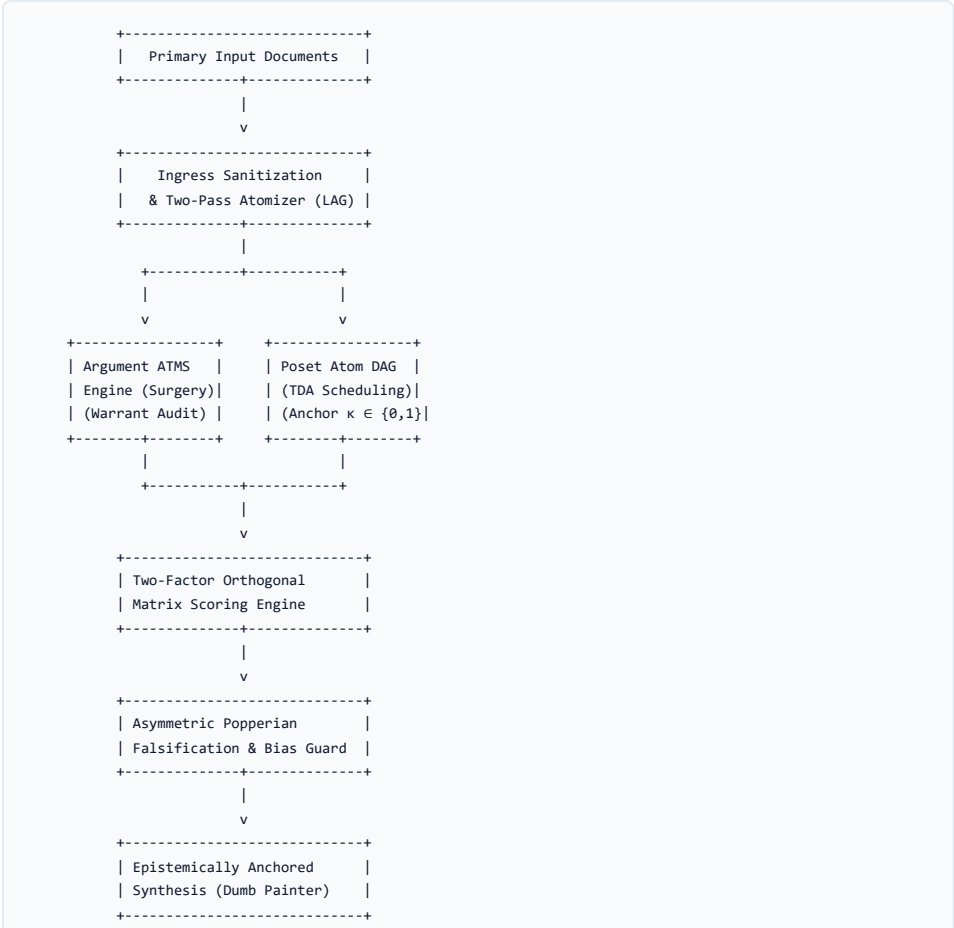
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3. Cognitive Directed Acyclic Graph (Cognitive DAG) Architecture

High-level cognitive transformations—ingress filtering, atomic decomposition, causal graph mining, matrix evaluation, falsification, and executive synthesis—are executed across an immutable, acyclic orchestration network:

$$\mathcal{G}_{\text{DAG}} = (\mathcal{V}_{\text{cog}}, \mathcal{E}_{\text{dep}}) \tag{6}$$

where $\mathcal{V}_{\text{cog}}$ represents typed computational operators and $\mathcal{E}_{\text{dep}}$ represents strict state and data dependencies.



3.1. Wave-Based Topological Execution

Because $\mathcal{G}_{\text{DAG}}$ is acyclic, Kahn's topological sort partitions $\mathcal{V}_{\text{cog}}$ into discrete, parallel computational waves $W_0, W_1, \ldots, W_{K-1}$:

$$W_0 = \{v \in \mathcal{V}_{\text{cog}} \mid \text{InDegree}(v) = 0\} \tag{7}$$

$$W_{k+1} = \left\{ v \in \mathcal{V}_{\text{cog}} \setminus \bigcup_{j=0}^{k} W_j \;\middle|\; \text{Parents}(v) \subseteq \bigcup_{j=0}^{k} W_j \right\} \tag{8}$$

Operators within wave W_k are executed concurrently as isolated asynchronous coroutines. Each operator $v \in W_k$ evaluates an invariant deterministic function over immutable Data Transfer Objects (DTOs):

$$\Phi_v : \prod_{u \in \text{Parents}(v)} \mathcal{O}_u \times \mathcal{S}_{\text{global}} \to \mathcal{O}_v \tag{9}$$

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4. Topological Directed Atom (TDA) Scheduling and Structural Argument Truth Maintenance

4.1. Topological Directed Atom (TDA) Poset Scheduling and Verification

Nomenclature Demarcation and Scope Calibration: To prevent terminological confusion with *Topological Data Analysis* (TDA in applied mathematics and machine learning, which studies persistent homology, simplicial complexes, and Betti numbers β_k over metric point clouds; Carlsson, 2009; Edelsbrunner & Harer, 2010), we explicitly clarify that in Quorum, the acronym **TDA** designates **Topological Directed Atoms** (and their associated dependency argumentation). The term *topological* is employed strictly in its foundational computer science and order-theoretic sense: Kahn's topological sorting (Kahn, 1962) on finite directed acyclic graphs inducing an Alexandroff topology and a specialization poset ($\mathcal{G}_{\text{LAG}}$). Quorum asserts no claims of computing algebraic homology groups or metric simplicial filtrations. Rather, TDA in Quorum denotes the **order-theoretic scheduling, exact lexical provenance, and cycle-free verification** of propositional claims:

1. **Strict Lexical Grounding:** Proposition truth values are grounded directly in verbatim source evidence. Unanchored assertions fail closed ($\kappa_i = 0$).
2. **Cycle Elimination and Poset Acyclicity:** Circular reasoning chains ($a_1 \implies a_2 \implies \ldots \implies a_1$) violate epistemic partial order invariants. Rather than invoking homological machinery, circularities are detected via Tarjan's strongly connected components algorithm (Tarjan, 1972) and isolated prior to evaluation, enforcing strict directed acyclicity across the proposition graph. Strongly connected components lacking an anchored exogenous premise are quarantined as circular epistemic loops.

4.1.1. Deterministic NLI/Entailment and Eradication of Second-Order Hallucination

A foundational failure mode in contemporary LLM evaluation (e.g., LLM-as-a-Judge, G-Eval, Self-RAG) is **Second-Order Hallucination** (*Quis custodiet ipsos custodes?*). If a secondary LLM is queried to determine whether a document entails a proposition ($P(\text{Entail} \mid \mathcal{D}, \tau_i)$), the verifier is itself an autoregressive stochastic sampler susceptible to sycophancy, confirmation bias, and the synthesis of *chimera quotes* (stitching isolated words across disparate paragraphs into a fabricated sentence).

To eliminate second-order hallucination, Quorum strips the LLM of judicial authority. Natural Language Inference (NLI) is decomposed into a **Tripartite Neurosymbolic Protocol**: the LLM operates strictly as a constrained syntactic sensor extracting proposition tuples, while propositional grounding and entailment are executed **exclusively in pure deterministic Python code** (`AnchorValidationService`):

1. **Gate 1: Physical Substring Invariant with Bidirectional Index Mapping:** The source text is stripped of HTML/formatting artifacts and normalized via Unicode NFKC and NFD diacritic decomposition. An invariant bidirectional index map

$\mathcal{M}_{\text{idx}} : \mathbb{N} \to \mathbb{N}$ maps normalized character indices back to original byte positions. The candidate quote $\mathcal{C}_i$ is validated via exact $O(N)$ containment:

$$\text{find}(\text{Norm}(\mathcal{C}_i), \text{Norm}(\mathcal{D})) \neq -1$$

If matched, the exact verbatim string is sliced from the original immutable text, preventing case-folding or formatting drifts from fabricating false evidence.

2. **Gate 2: Entropy Gate & Contiguity Guard (Chimera Quote Defense):**
   - *Entropy Gate (< 10 chars):* Short quotes under 10 characters (e.g. isolated buzzwords) are strictly forbidden from fuzzy matching. 100% exact substring containment is required.
   - *Contiguity Guard (< 30 chars):* Short phrases enforce strict contiguous sequence matching (`fuzz.partial_ratio`). This prevents the model from picking disparate words from different pages and presenting them as a continuous quote.
   - *Safety Ceiling (> 1000 chars):* Quotes exceeding 1000 characters trigger an immediate `SemanticEvidenceError` fail-fast crash to prevent prompt injection and quote stuffing.

3. **Gate 3: Cross-Speaker Provenance Boundary Guard:** Discourse is segmented into XML-fenced provenance capsules: `<user_payload>` for human supervisor inputs and `<ai_draft_context>` for model drafts. When evaluating human capability ($\sigma_i = \text{USER}$), candidate quotes must reside strictly inside `<user_payload>`:

$$\text{Prov}(\mathcal{C}_i, \sigma_i) = \mathbb{I}\left(\text{find}(\text{Norm}(\mathcal{C}_i), \text{Norm}(\mathcal{D}_{\text{speaker}})) \neq -1\right)$$

   If an LLM quotes the AI assistant's draft to substantiate a human competence claim, the verifier halts with a `PROVENANCE_VIOLATION` error, mathematically preventing cross-speaker competence pollution.

4. **Gate 4: Reasoning Trace Triangulation:** The model's Pydantic response requires an explicit reasoning trace $\mathcal{T}_i$ specifying an exact syntactic anchor (`[2. SYNTACTIC ANCHOR: "..."]`) and decision (`[5. VALIDATION DECISION: PASS/FAIL]`). Pure Python triangulation asserts three consistency invariants:
   - *Contradiction Ban:* If $\mathcal{T}_i$ concludes `FAIL`, but quote $\mathcal{C}_i$ is populated, the atom is rejected for logical contradiction.
   - *Empty Anchor Ban:* If $\mathcal{T}_i$ declares `SYNTACTIC ANCHOR: NONE`, unanchored extraction fails closed.
   - *Hallucinated Anchor Ban:* The anchor string parsed from the trace must physically exist in the source document; fabricated anchors trigger immediate rejection.

5. **Gate 5: Calibrated Zero-Trust Strictness Curve:** Under Quorum's continuous strictness parameter $\theta \in [0, 100]$, fuzzy matching thresholds scale monotonically. At $\theta \geq 100$ (Absolute Zero-Trust), fuzzy fallback is completely disabled, enforcing 100% exact byte matching.

Under this five-gate architecture, propositional grounding does not rely on LLM self-evaluation. An atom τ_i is accepted ($\kappa_i = 1$) if and only if all five deterministic Python gates succeed, bounding the false acceptance probability by the physical string collision

rate $\epsilon_{\text{lexical}} \to 0$.

Formal Scope and the Semantic Undecidability Bound: We explicitly distinguish between **Physical Citation Provenance** and **Universal Semantic Entailment**. In unconstrained natural language, determining whether an arbitrary premise $\mathcal{C}$ semantically entails an arbitrary conclusion τ without human supervision is undecidable (Rice's Theorem applied to semantic properties of formal discourse). Quorum does not claim universal automated semantic proof over arbitrary text. Instead, Quorum solves the practical enterprise failure mode: it bounds hallucination risk by enforcing that propositions τ_i originate from human-curated, pre-compiled BARS rubrics, and constrains the stochastic model to locating exact physical evidence spans whose grammatical anchors pass deterministic Python verification.

4.2. Argumentative Structural Dependency Models (A-SCM) and Truth Maintenance

Ontological Scope and the Enthymeme Boundary: Unlike physical systems where variables $\mathbf{V}$ represent empirical data-generating mechanisms, natural discourse is inherently pragmatic, elliptical, and enthymematic (Toulmin, 1958; Grice, 1975). Human authors routinely omit common-sense warrants and intermediate mechanisms. Therefore, Quorum does not claim that textual analysis discovers objective physical causality. Instead, Quorum formalizes the **Argumentative Structural Causal Model (A-SCM)**, evaluating the *internal explanatory sufficiency and structural evidentiary rigor* of the authored thesis.

We formalize the discourse argument structure through the five-tuple A-SCM:

$$\mathcal{M}_{\text{arg}} = \langle \mathbf{U}, \mathbf{V}, \mathbf{W}, \mathbf{F}, P(\mathbf{U}) \rangle \tag{10}$$

where $\mathbf{V} = \{V_1, \dots, V_n\}$ represents explicit propositional claims grounded in verbatim text ($\kappa = 1$), $\mathbf{W}$ represents explicit Toulmin warrants (backing and explanatory transmission mechanisms), $\mathbf{U} = \{U_{\text{enthymeme}}, U_{\text{context}}\}$ represents unstated background assumptions and latent conversational confounders, and $\mathbf{F} = \{f_1, \dots, f_m\}$ defines structural argument dependencies:

$$V_i = f_i(\mathbf{PA}_i, \mathbf{W}_i, U_i), \quad \mathbf{PA}_i \subseteq \mathbf{V} \setminus \{V_i\} \tag{11}$$

Operationalization of Structural Interventions via Discrete Graph Surgery & Truth Maintenance: In structural causal modeling (Pearl, 2009, Chapter 3.2), an intervention on variable X is defined graph-theoretically by replacing its structural equation with a constant assignment $X = x^*$, which corresponds to severing all incoming parent edges in the surgery graph $\mathcal{G}_{\overline{X}}$.

A foundational question in NLP argumentation is: *Which computational system physically evaluates the downstream consequences of this structural intervention in discourse?* If a secondary LLM is re-prompted to judge whether claim Y survives without premise X, the pipeline relapses into second-order stochasticity and confirmation bias.

In Quorum, structural intervention is **NOT executed via LLM prompts or statistical regression**. The extracted discourse is compiled into a discrete Argument DAG $\mathcal{G}_{\text{LAG}} = (\mathcal{V}_{\text{atom}}, \mathcal{E}_{\text{deps}})$, and interventions are evaluated via **deterministic graph surgery and Assumption-Based Truth Maintenance Systems (ATMS)** (Doyle, 1979; de Kleer, 1986; Dung, 1995) implemented in pure Python (`TopologicalEvaluator` and `CausalDiscoveryEngine`), formalized in Algorithm 1:

```
Algorithm 1: Discrete Structural Argument Surgery & Truth Maintenance (ATMS)
================================================================================
Input  : Argument DAG G = (V, E, W), perturbed node X in V, counterfactual state x* in {FAILED, BLOCKED}
Output : Perturbed state mapping S*, Reachability Cascade Reach(X), Root Causes RC
================================================================================
1: S* <- InitialStateMap(V)                          // Base status from lexical grounding Gate
2: E_cut <- E \ { (P, X) | P in Parents(X) }         // Sever incoming parent edges (Structural graph
surgery)
3: S*[X] <- x*                                       // Clamp intervened node to counterfactual state
4: Queue Q <- [ X ]
5: Visited Vis <- { X }
6: Cascade <- []
7: while Q is not empty do
8:     curr <- Q.pop_front()
9:     for each child C in Children(curr, E_cut) do
10:        if C not in Vis then
11:            // Warrant Independence Gate: Check if an alternative valid warrant exists
12:            has_alternative_warrant <- exists P' in Parents(C, E_cut) \ Vis
13:                                       such that S*[P'] == PASSED and kappa(W_{P'C}) == 1
14:            if not has_alternative_warrant then
15:                S*[C] <- BLOCKED
16:                Vis.add(C)
17:                Cascade.append(C)
18:                Q.push_back(C)
19:            end if
20:        end if
21:    end for
22: end while
23: RC <- { v in V | S*[v] in {FAILED, BLOCKED} and BlameParents(v) == emptyset }
24: return (S*, Cascade, RC)
================================================================================
```

Algorithmic Guarantees of Algorithm 1: 1. **Zero Second-Order LLM Calls:** Lines 7–22 operate in linear deterministic time $\mathcal{O}(|\mathcal{V}| + |\mathcal{E}|)$ over in-memory adjacency lists in pure Python. No stochastic LLM sampling occurs during perturbation evaluation. 2. **Deterministic Edge Severing (Structural Graph Surgery):** Line 2 executes structural graph surgery ($\mathcal{G}_{\overline{X}}$) by pruning incoming edges in the adjacency structure while holding invariant the downstream topology. 3. **Assumption-Based Truth Maintenance & Toulmin Warrant Protection:** Lines 12–13 implement assumption-based justification checking (de Kleer, 1986). Dependent claim C survives if supported by an alternative valid warrant $\mathbf{W}_{P'C}$ backed by an unperturbed parent ($\kappa = 1$). If no valid alternative warrant exists, belief in C is retracted, deterministically transitioning dependent nodes to `BLOCKED`. 4. **Root Cause Attribution:** Line 23 uniquely identifies primary epistemic breakdown points, differentiating root premise failures from downstream cascaded blocks without penalizing authors repeatedly for secondary effects.

Scope Disambiguation (Discrete ATMS Surgery vs. Observational Do-Calculus): We explicitly repudiate any rhetorical conflation between **Discrete Structural Argument Surgery** and **Observational Confounding Adjustment via Pearl's Three Rules of Do-Calculus**. The classical do-calculus (Pearl, 1995, 2009) is an axiomatic apparatus designed for non-parametric identification of interventional probability distributions $P(Y \mid do(X))$ from non-experimental observational joint distributions $P(V)$ in the presence of unobserved

statistical confounders. In Quorum's discrete text analysis, there are no observational probability distributions, random variables, or backdoor adjustment summations. Rather, Quorum utilizes Pearl's graph-theoretic definition of an intervention ($\mathcal{G}_{\overline{X}}$: severing incoming dependencies) strictly as an operational counterpart to de Kleer's (1986) premise retraction in an Assumption-Based Truth Maintenance System (ATMS). Algorithm 1 deterministically audits whether an author's conclusions possess independent evidentiary warrants ($\mathbf{W}$) or experience justification collapse when antecedent premises are invalidated.

This maps structurally to the epistemic demands of the Three Rungs of Argument Rigor: * **Rung 1 (Associative Co-occurrence & Fluff):** Unidirectional linguistic correlation lacking explicit warrants ($\mathbf{W} = \emptyset$). Listing co-occurring industry trends without causal transmission mechanisms incurs severe passive delegation penalties. * **Rung 2 (Intervention Audit & Warrant Isolation):** Simulating structural interventions by pruning parent dependencies. The audit evaluates whether the text explicitly defines operational mechanisms $\mathbf{W}_{XY}$ isolating the claim from common-cause confounding (U_{XY}). * **Rung 3 (Counterfactual Stress-Testing):** Evaluating counterfactual boundary transitions. The audit examines whether the author subjected proposed strategies to "what-if" boundary conditions, failure thresholds, negative externalities, and resource exhaustion scenarios.

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5. Two-Factor Orthogonal Matrix Scoring and Asymmetric Falsification

5.1. BARS Criteria Weighting and Division-by-Zero Protection

Evaluation matrices implement Behaviorally Anchored Rating Scales (BARS) distributed across five hierarchical tiers $l \in \{1, \ldots, 5\}$ with natural weight function $w(l) = l$. Every atom $a_i \in \mathcal{A}$ resolves to a discrete status:

$$\omega(a_i) \in \{\text{PASSED}, \text{FAILED}, \text{DLQ}\} \tag{12}$$

where DLQ (Dead-Letter Queue) denotes unresolvable processing exceptions.

For positive capability atoms ($\pi_i = +1$), achieved points A and effective maximum points M are computed over all tiers:

$$A = \sum_{l=1}^{5} l \cdot h_l, \quad M = \sum_{l=1}^{5} l \cdot (t_l - d_l) \tag{13}$$

where h_l is passed hits, t_l is total criteria count, and d_l is DLQ exceptions at tier l. The weighted linear ratio ρ is safeguarded against division-by-zero singularities:

$$\rho = \begin{cases} \frac{A}{M}, & M > 0 \\ 0.0, & M \leq 0 \end{cases} \tag{14}$$

5.2. Asymmetric Popperian Falsification Multiplier

Negative anti-pattern atoms ($\pi_i = -1$) act as non-additive gatekeeping falsifiers. If an anti-pattern violation is confirmed ($\omega(a_i) = \text{FAILED}$ relative to clean performance), a penalty deduction $p_k \in (0, 1)$ is registered. The composite multiplier is formed multiplicatively:

$$\Pi_{\text{penalty}} = \begin{cases} \max\left(1 - P_{\max},\ \prod_{k=1}^{K}(1 - p_k)\right), & K > 0 \\ 1.0, & K = 0 \end{cases} \tag{15}$$

where $P_{\max} = 0.40$ is the hard-coded ceiling capping total cumulative deductions at 40%.

5.3. Continuous Piecewise Power-Curve Damping

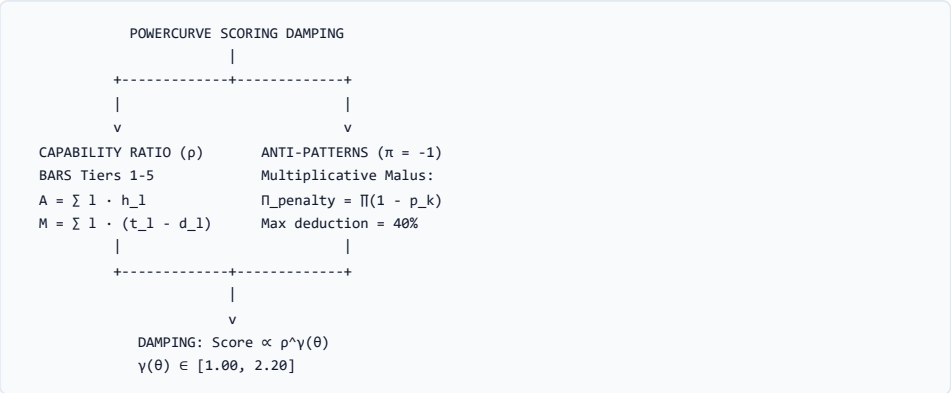
To penalize performance clustering at trivial lower tiers ($l \in \{1, 2\}$), the system maps user strictness $\theta \in [0, 100]$ to a damping exponent $\gamma(\theta) \in [1.00, 2.20]$ via continuous piecewise linear interpolation:

$$\gamma(\theta) = \begin{cases} 1.00, & \theta \le 0 \\ 1.00 + \left(\frac{\theta}{50}\right) \cdot 0.25, & 0 < \theta \le 50 \\ 1.25 + \left(\frac{\theta - 50}{35}\right) \cdot 0.45, & 50 < \theta \le 85 \\ 1.70 + \left(\frac{\theta - 85}{15}\right) \cdot 0.50, & 85 < \theta \le 100 \\ 2.20, & \theta > 100 \end{cases} \tag{16}$$

The final mapped score across the range $[S_{\min}, S_{\max}]$ is calculated as:

$$\text{Score}(\rho) = \begin{cases} S_{\min} + \left(\rho^{\gamma(\theta)} \cdot \Pi_{\text{penalty}}\right) \cdot (S_{\max} - S_{\min}), & M > 0 \\ S_{\min}, & M \le 0 \end{cases} \tag{17}$$

Mathematical Property 1 (Score Monotonicity and Lipschitz Stability): Continuous power-curve damping is an architectural design choice formulated to eliminate the "cliff-edge failure" characteristic of discrete zero-trust gates (where a single missing token triggers a discontinuous score collapse). From an analytical standpoint, the scoring function exhibits two key mathematical properties for any calibrated exponent $\gamma \ge 1.0$ and penalty multiplier $\Pi_{\text{penalty}} \in (0, 1]$: 1. **Monotonicity:** $\frac{\partial \text{Score}}{\partial \rho} = \gamma \rho^{\gamma - 1} \Pi_{\text{penalty}}(S_{\max} - S_{\min}) \ge 0$ everywhere on $[0, 1]$. Validating an additional capability atom monotonically improves or maintains total score. 2. **Lipschitz Stability:** Because $\frac{\partial^2 \text{Score}}{\partial \rho^2} \ge 0$, the maximum rate of change is attained at the upper boundary $\rho = 1$, yielding the optimal Lipschitz constant $K_L = \gamma \Pi_{\text{penalty}}(S_{\max} - S_{\min})$. By the Mean Value Theorem, $|\text{Score}(\rho_1) - \text{Score}(\rho_2)| \le K_L |\rho_1 - \rho_2|$, ensuring that bounded evidence noise cannot trigger unbounded diagnostic volatility.



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6. Architectural Principles and Verified Mathematical Properties

Epistemic Calibration and System Scope: To preserve strict scientific integrity and avoid theoretical inflation, we explicitly partition the formal framework into two practical categories: 1. **Architectural Principles:** Structural design invariants inherited from established computer science (Church-Rosser lambda confluence, Kahn Process Networks, and discrete topological DAG scheduling) that govern deterministic asynchronous orchestration. 2. **Mathematically Verified System Properties:** Analytical properties of our verification and scoring pipelines (lexical error suppression bounds, Lipschitz score stability, and topological reachability) that guarantee numerical and epistemic predictability.

Architectural Principle 1: Confluent Neurosymbolic Wave Scheduling

Architectural Principle 1 (Wave Confluence under Kahn Ordering and Pure Computations). *Let $\mathcal{G}_{\text{DAG}} = (\mathcal{V}_{\text{cog}}, \mathcal{E}_{\text{dep}})$ be a finite directed acyclic graph partitioned into K disjoint topological waves $W_0, W_1, \ldots, W_{K-1}$ via Kahn's partial ordering (Kahn, 1962). Let each node operator $\Phi_v : \prod_{u \in \text{Parents}(v)} \mathcal{O}_u \times \mathcal{S}_{\text{global}} \to \mathcal{O}_v$ be a pure compute function executing over immutable, validated Data Transfer Objects (DTOs) with zero shared memory side-effects. By the Church-Rosser theorem (Church & Rosser, 1936) and the determinism invariants of Kahn Process Networks (Kahn, 1974), any arbitrary asynchronous resolution order or coroutine scheduling permutation $\pi \in \text{Perm}(W_k)$ within `asyncio.TaskGroup` converges deterministically to the identical unique wave state snapshot $\mathcal{O}_{W_k}$, and the global system pipeline terminates at an invariant fixed-point state $\mathcal{S}^* = (\Phi_{W_{K-1}} \circ \cdots \circ \Phi_{W_0})(\mathcal{S}_0)$.*

Architectural Context & Contribution: Confluence of pure functions over immutable inputs in acyclic topologies is a well-established foundational property of theoretical computer science (Church & Rosser, 1936; Kahn, 1974). We do not claim wave confluence as a novel mathematical discovery. Rather, Quorum's real-world contribution is the **strict architectural enforcement of referential transparency within a distributed LLM runtime**: 1. Operators communicate exclusively via immutable, frozen Pydantic DTOs (`ConfigDict(frozen=True)`). 2. Cognitive engines adhere to the Pure Compute Engine Law (100% computational pipelines with zero shared locks, event listeners, or mutable

globals). 3. Concurrency throttling is cleanly decoupled into dynamic HTTP semaphore pools, preventing runtime deadlocks while guaranteeing bit-level deterministic state reproduction across re-executions.

Architectural Property 1: Deterministic Error Suppression via Lexical Grounding and Role Asymmetry

Property 1 (Deterministic Epistemic Error Suppression). *Let an ensemble of M frontier LLM evaluators exhibit pairwise error correlation $\mathrm{Cov}(X_i, X_j) = \rho_{ij} \geq \rho_{\min} > 0$ due to shared pretraining corpora (Common Crawl, Wikipedia). While democratic majority voting hits an asymptotic error floor governed by effective sample size $M_{\text{eff}} \leq 1/\rho_{\min}$, Quorum's two-tier neurosymbolic firewall—combining Role-Asymmetric Popperian Falsification ($\pi = -1$) with deterministic Physical Lexical Grounding ($\kappa \in \{0, 1\}$) over pre-compiled BARS criteria—decouples epistemic error suppression from model independence, bounding system acceptance error strictly by exact lexical substring collision tolerances:*

$$P(\text{System False Accept}) \leq P_{\text{prior}}(\text{Hallucination}) \cdot P(\kappa_i = 1 \mid \text{Hallucination}) \leq \epsilon_{\text{lexical}} \rightarrow$$

Analytical Verification & Derivation:

1. **Failure of Democratic Consensus under Correlated Priors:** Let $X_1, \ldots, X_M \in \{0, 1\}$ be binary error indicators for model evaluators on a deceptive prompt. Due to overlapping pretraining distributions, the average pairwise correlation is non-zero: $\bar{\rho} \geq \rho_{\min} > 0$. The effective sample size is bounded asymptotically:

$$M_{\text{eff}} = \frac{M}{1 + (M-1)\bar{\rho}} \implies \lim_{M \to \infty} M_{\text{eff}} = \frac{1}{\bar{\rho}} < \infty \tag{19}$$

If the prompt triggers a shared pretraining attractor where individual error probability $p > 0.5$, democratic voting inverts: $P(\text{Consensus Error}) \to 1$ as $M \to \infty$. Thus, model multiplexing alone cannot guarantee epistemic reliability.

2. **Asymmetric Popperian Falsification Barrier:** Quorum breaks symmetric voting by establishing role asymmetry. An explicit adversarial falsifier is parameterized with negative polarity ($\pi_i = -1$) under the Popperian Null Hypothesis: a claim is assumed *invalid or pseudo-competent* unless surviving rigorous counter-evidence extraction. By separating the extraction agent from the falsification auditor, the joint acceptance condition requires:

$$P(\text{Claim Pass}) = P(\text{Extracted}) \cdot (1 - P(\text{Falsified} \mid \text{Extracted})) \tag{20}$$

3. **Deterministic Physical Lexical Firewall ($\kappa \in \{0, 1\}$):** Regardless of LLM consensus or correlation $\bar{\rho}$, every extracted claim must satisfy the strict lexical grounding predicate:

$$\kappa_i = \mathbb{I}\left(\text{find}(\text{Norm}(\mathcal{C}_i), \text{Norm}(\mathcal{D})) \neq -1\right) \in \{0, 1\} \tag{21}$$

where $\mathcal{C}_i$ is the exact textual citation and $\mathcal{D}$ is the raw immutable source document. A hallucinated assertion a_{halluc} can be accepted if and only if its verbatim citation physically matches an existing text span in $\mathcal{D}$:

$$P(\text{System False Accept}) = P(\text{Hallucinated Assertion}) \cdot P(\text{find}(\mathcal{C}_{\text{halluc}}, \mathcal{D}) \neq -1) \quad (22)$$

Because the lexical matcher is an exact, non-fuzzy string boundary invariant implemented in deterministic Python code, $P(\text{find}(\mathcal{C}_{\text{halluc}}, \mathcal{D}) \neq -1) \leq \epsilon_{\text{lexical}}$, where $\epsilon_{\text{lexical}}$ is the probability of random string collision across the finite source corpus length $|\mathcal{D}|$. Furthermore, because target propositions τ_i are constrained to pre-compiled BARS rubrics rather than unconstrained model generation, semantic drift is bounded by rubric specification. Even when model representations are fully correlated ($\bar{\rho} \to 1$), hallucination acceptance is suppressed deterministically to zero.

Architectural Property 2: Discrete ATMS Reachability and Cascade Fault Isolation

Property 2 (Warrant Independence and Cascade Fault Isolation). *Let an Argumentative Discourse SCM be compiled into an explicit discrete argument DAG $\mathcal{G}_{\text{LAG}} = (\mathcal{V}_{\text{atom}}, \mathcal{E}_{\text{deps}})$, where explicit claims $\mathbf{V}$ and operational warrants $\mathbf{W}$ are grounded in text ($\kappa = 1$). Assume latent conversational enthymemes $U_{\text{enthymeme}}$ act as potential unstated confounders. If a discrete structural graph surgery $\text{do}(X = \text{FAILED})$ on the surgery graph $\mathcal{G}_{\overline{X}}$ alters dependent claim validity via deterministic reachability $\text{Reach}(\mathcal{G}_{\overline{X}}, X) \cap \{Y\} \neq \emptyset$, while the reverse perturbation leaves antecedent truth invariant $X \notin \text{Reach}(\mathcal{G}_{\overline{Y}}, Y)$, and the text provides an explicit physical warrant $\mathbf{W}_{XY}$, then the directed argumentative dependency $X \xrightarrow{\mathbf{W}_{XY}} Y$ is structurally sound, self-contained, and shielded from enthymemic confounding.*

Topological Derivation & System Mechanics:

1. **Discrete Graph Surgery via Truth Maintenance:** Rather than attempting observational density estimation via statistical do-calculus, discourse graph surgery is evaluated deterministically in pure Python (`TopologicalEvaluator` and `CausalDiscoveryEngine`). Applying structural intervention $\text{do}(X = \text{FAILED})$ severs incoming edges $\mathbf{PA}_X \to X$ in the adjacency structure ($\mathcal{G}_{\overline{X}}$) and clamps X's state to `FAILED`, triggering ATMS label retraction across downstream dependencies.

2. **Topological Asymmetry and Reachability:** Let $\text{Reach}(\mathcal{G}, v)$ denote the set of all nodes reachable via directed paths from v. The invariant $X \notin \text{Reach}(\mathcal{G}_{\overline{Y}}, Y)$ proves that premise X is topologically upstream from outcome Y:

$$(X \perp_{\mathcal{G}_{\overline{Y}}} Y) \implies Y \notin \text{Ancestors}(X) \quad (23)$$

No backward causal flow exists in the authored discourse. Concurrently, $Y \in \text{Reach}(\mathcal{G}_{\overline{X}}, X)$ confirms that Y is structurally downstream from X:

$$(Y \not\perp_{\mathcal{G}_{\overline{X}}} X) \implies X \in \text{Ancestors}(Y) \quad (24)$$

3. **Enthymemic Confounder Isolation via Grounded Warrants:** Suppose an unstated assumption $U_{\text{enthymeme}}$ creates a spurious correlation $X \leftarrow U_{\text{enthymeme}} \to Y$ (a rhetorical leap where the author omitted mediating operational steps). Under the discrete intervention $\text{do}(X = \text{FAILED})$, the path through X is severed. If the author omitted explicit warrants ($\mathbf{W}_{XY} = \emptyset$), the topological fault evaluator detects that:

$$\text{BlameParents}(Y) = \{X\} \implies \text{Status}(Y) \leftarrow \text{BLOCKED} \tag{25}$$

and the audit flags an ungrounded argumentative leap (Rung 1).

Conversely, if the author explicitly stated and grounded the operational warrant $\mathbf{W}_{XY}$ ($\kappa(\mathbf{W}_{XY}) = 1$), the active transmission mechanism establishes an independent verifiable edge $X \xrightarrow{\mathbf{W}_{XY}} Y$. In discrete graph surgery, the warrant $\mathbf{W}_{XY}$ provides the deterministic transmission mechanism, shielding the claim from unstated enthymemes and confirming Rung 2/3 explanatory rigor.

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7. Error Propagation and Cascade Suppression Pipeline

Unlike standard unconstrained LLM pipelines where intermediate hallucinations cascade unchecked, Quorum forces state through a four-stage suppression barrier:

$$\mathcal{S}_{\text{out}} = (\Phi_{\text{synth}} \circ \Phi_{\text{score}} \circ \Phi_{\text{fals}} \circ \Phi_{\text{TDA}} \circ \Phi_{\text{ingress}})(\mathcal{S}_{\text{in}} + \xi) \tag{26}$$

where ξ represents stochastic semantic noise.



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8. Empirical Validation: The Sitra Strategic Foresight Audit

To benchmark the theoretical framework against real-world data, the Quorum engine was deployed to audit an executive foresight analysis task based on the Sitra Megatrend dataset (2017–2023). The audit evaluated both the supervisory interaction (`Chat`) and the produced strategic brief (`Lopputuote`) under strictness $\theta = 50\%$ ($\gamma = 1.25$).

8.1. Evaluated Score Breakdown and Reconciliation

Parameter	HACM (Chat)	Causal (Report)
Execution ID	exe_c0953f62f2014b40	exe_c0953f62f2014b40
Total Criteria Evaluated	25	25
Anti-Pattern Checks ($\pi = -1$)	13/13 clean	13/13 clean
Penalty Multiplier (Π_{penalty})	**1.0000**	**1.0000**
Capability Criteria ($\pi = +1$)	12 total	12 total
Level 1 Passed (Weight x1)	2 / 2 passed (2.0)	1 / 1 passed (1.0)
Level 2 Passed (Weight x2)	1 / 2 passed (2.0)	2 / 2 passed (4.0)
Level 3 Passed (Weight x3)	1 / 2 passed (3.0)	0 / 3 passed (0.0)
Level 4 Passed (Weight x4)	0 / 3 passed (0.0)	0 / 3 passed (0.0)
Level 5 Passed (Weight x5)	0 / 3 passed (0.0)	0 / 3 passed (0.0)
Achieved Points (A)	**7.0 pts**	**5.0 pts**
Maximum Points (M)	**39.0 pts**	**41.0 pts**
Linear Ratio ($\rho = A/M$)	17.95% $(7/39)$	12.20% $(5/41)$
Exponent (γ at 50%)	1.2500	1.2500
Curved Ratio ($\rho^{1.25}$)	0.116827	0.072066
Final Normalized Score	**11.68 / 100 (11.7%)**	**7.21 / 100 (7.2%)**
Verdict	Sparrer Profile	Associative Rung 1

8.2. Granular Causal Graph Extraction Metrics

The automated pipeline parsed the corpus into a Linked Atom Graph containing **255 extracted causal propositions**:

Metric Parameter	Value	Architectural Significance
Total Causal Propositions	255	Raw Cognitive Proposition Pool
Validated Connected Nodes	161 (63.1%)	Context-Anchored Evidence Chains
Orphan Claims ($\text{InDeg} = 0$)	94 (36.9%)	Unanchored Leaps or Exogenous Axioms
Maximum In-Degree (Hub)	5.0	Central Terminal Recommendation
Mean In-Degree Connectivity	0.41	Sparse Linear Argumentative Topology

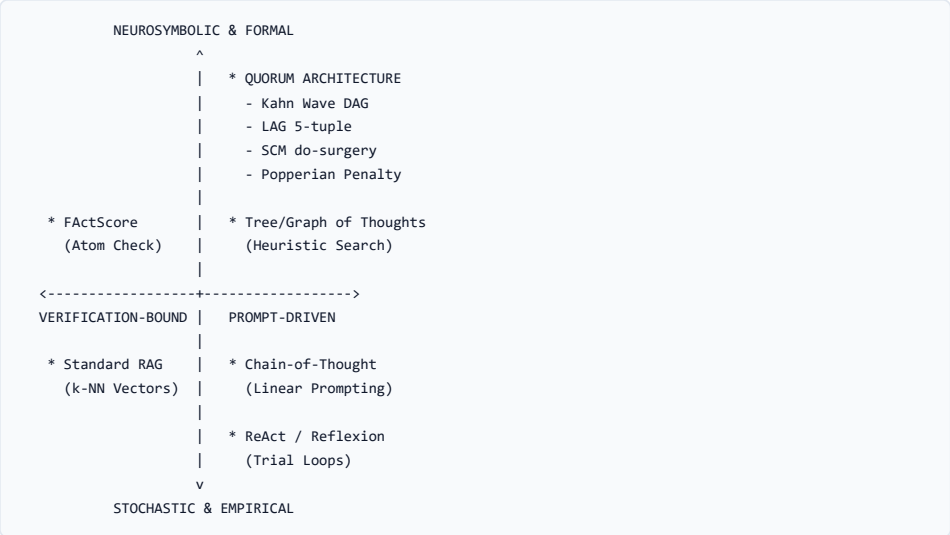
8.3. Cognitive Depth and the "Illusion of Control"

Linguistic analysis revealed a severe disparity between form and substance:

- **Mechanical Phrase Load:** $0.00/2.0$ (Zero clichés or empty buzzwords detected).
- **Cognitive Authenticity:** $1.47/3.0$ (Low, unreflective routine delegation).
- **Form-to-Content Disparity Ratio:** 1.53 (Verdict: **IN CONFLICT**).

The user formulated concise, jargon-free prompts (*"mikä on sitran näkemys suunnasta eteenpäin ei siis toivetila"*), which created an **Illusion of Control**. The operator assumed the concise prompt guaranteed an analytically rigorous output. In reality, the AI's first associative draft was accepted without counterfactual testing, sensitivity analysis, or trade-off evaluation.

9. Synthesis: Quorum's Positioning in the Research Landscape



9.1. Comparison with Reasoning and Orchestration Frameworks

- **Chain-of-Thought (CoT) (Wei et al., 2022):** CoT relies entirely on the autoregressive distribution to maintain reasoning consistency. As demonstrated in Equation (2), early errors cascade exponentially. Quorum eliminates autoregressive reasoning chains, substituting them with Kahn-sorted wave execution (Architectural Principle 1) over an explicit acyclic network.
- **Tree of Thoughts (ToT) (Yao et al., 2023) and Graph of Thoughts (GoT) (Besta et al., 2024):** ToT and GoT structure thoughts into search graphs (BFS/DFS). However, their nodes remain unstructured natural language strings, and state pruning relies on LLM self-evaluation. Quorum's nodes are strictly typed computational operators producing validated DTOs, and state termination is mathematically guaranteed.
- **ReAct (Yao et al., 2022) and Reflexion (Shinn et al., 2023):** Reflexion implements heuristic self-correction loops. However, recent empirical work (Huang et al., 2023) proves that LLMs cannot reliably self-correct without external deterministic ground truth. Quorum replaces circular self-reflection with external, asymmetric Popperian falsification.

9.2. Comparison with Retrieval and Factuality Verification

- **Standard RAG (Lewis et al., 2020) and Self-RAG (Asai et al., 2023):** Standard RAG retrieves passage chunks via embedding similarity (k-NN), which introduces irrelevant context and fails on fine-grained logic. Quorum rejects chunk-level vector retrieval, deploying the epistemic atomization morphism $\alpha(\mathcal{D}) \to \mathcal{A}$ to evaluate discrete propositions against an absolute lexical null hypothesis.
- **FActScore (Min et al., 2023) and RARR (Gao et al., 2023):** FActScore decomposes text into atomic facts, but evaluates them symmetrically as binary factual counters. Quorum structures atoms into the five-tuple $a_i = \langle \tau_i, \mathcal{C}_i, \sigma_i, \pi_i, \kappa_i \rangle$, tracking speaker attribution, evidentiary polarity, and directional causality within a coherent multigraph.

9.3. Cross-Paradigm Architectural Comparison

Dimension	Standard RAG / CoT	ReAct / Reflexion	Quorum Engine
Orchestration	Linear Sequence	Heuristic Loop	**Kahn Wave DAG**
State Verify	None (Generative)	Self-Reflection	**DTO Invariants**
Grounding	k-NN Dense Vector	Tool Output	**Lexical Null H_0**
Causal Logic	None (Spurious)	Trial-and-Error	**A-SCM & Truth Maint.**
Scoring	LLM-as-a-Judge	Heuristic	**Lipschitz BARS**
Falsification	None	Reranking	**Asymmetric Malus**
Termination	Max Length	Max Iterations	**Proved Acyclic**
Agency Audit	None	None	**HACM Tracking**

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10. Limitations and Practical Considerations

1. **Computational Overhead and FinOps Allocation:** Multi-provider consensus voting and deterministic lexical verification (Architectural Property 1) combined with counterfactual branch expansion increase inference token costs. In production enterprise deployments, Quorum mitigates this through selective routing: lightweight models execute initial atomization, while heavy multi-provider ensembles are reserved for critical Level 4–5 causal nodes.
2. **Exogenous Axiom Misattribution:** In sparse strategic corpora, separating genuine exogenous axioms (U_i) from unanchored orphan claims relies on accurate bounding box parsing. If an author introduces implicit domain premises without citations, TDA provenance verification flags them as orphans ($\kappa_i = 0$). Domain-specific white-listing of axiomatic priors resolves this friction.
3. **Cohort Percentile Calibration:** The scores reported (11.7% and 7.2%) reflect strict BARS damping ($\gamma = 1.25$). Without peer comparison, operators may perceive these scores as excessively punitive. Providing percentile rankings against normalized organizational cohorts contextualizes the diagnostic.

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11. Conclusions

The Quorum architecture establishes a mathematically rigorous foundation for turning stochastic large language models into verifiable, epistemically anchored expert systems. By replacing unconstrained autoregressive generation with:

- Kahn wave-based DAG scheduling (Architectural Principle 1),
- Deterministic physical lexical grounding and role-asymmetric Popperian falsification (Architectural Property 1),
- Monotonic, Lipschitz-stable BARS power-curve scoring (Mathematical Property 1), and
- Discrete structural argument surgery and ATMS-based fault isolation (Architectural Property 2),

the framework eliminates hallucination cascades and enforces structural accountability. The Sitra empirical audit validates that Quorum cuts through polished surface fluency to diagnose cognitive delegation traps, providing an auditable, verifiable methodology for enterprise and strategic artificial intelligence.

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